10102022

VCD/ SYIT SEM III DATABASE MANAGEMENT SYSTEM 75 MARKS 2½ HOURS

N.B. (1) All questions are **compulsory**.
(2) Figures to the right indicate marks.
(3) Answers to the same question must be written together.
(4) Mixing of sub-questions is not allowed.

Q1. Attempt any three of the following. **15**

a) Explain network model.
b) Explain the different types of Database Users.
c) List and explain building blocks of data model.
d) Describe with example the concept of Mapping Cardinalities.
e) List and Explain Codd's Rules (any five).
f) A football club has a name and ground and is made up of players. A player can play for only one club and a manager, represented by his name manages a club.
A footballer has a registration number, name and age. A club manager also buys players. Each club plays against each other club in the league and matches have a date, venue and score.

Q2. Attempt any three of the following. **15**

a) Explain the Normalization with its forms.
b) Write a note on relational algebra.
c) What is the tuple relational calculus? Explain it.
d) Differentiate between Relational Algebra and Relational Calculus.
e) Explain set operator with suitable example.
f) Write a short note on Domain Relational Calculus.

Q3. Attempt any three of the following. **15**

a) Explain correlated Subqueries with its types.
b) Explain Group by clause and Order by clause with example.
c) Write the concept of views.
d) Explain aggregate functions with example
e) Create student table (s_id, s_fname, s-lname, add, stream
f) consider employee table (e_id, e_name, e_add, e_city, e_dept, e_sal).
Solve the following query.
1) Display the total number of employees.
2) Display the names of the employees located in city 'Pune'.
3) Display the employee's name whose name begin with 'S'.
4) Display the employee's name whose salary is greater than 10000.
5) Display all the employees located in each city.

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Q4. Attempt any three of the following. 15

a) Explain properties of transaction.
b) Write a short note on view serializability.
c) What are concurrent transactions? Explain.
d) Write a short note on 2PL.
e) Discuss various strategies used for handling deadlock.
f) Write a short note on transaction state.

Q5. Attempt any three of the following. 15

a) Create a function to swap numbers.
b) Explain while loop statement in PL/SQL block with an example.
c) How to create and execute store procedure?
d) Create a trigger to insert the name of the student in UPPERCASE in the table student (rno, name).
e) Explain CASE statement with example.
f) What do you mean by parameterized cursor? How to declare it? Explain.